

Assignment 2
IB3702 Mathematics for Machine learning

Harman Singh

November 7, 2025

Problem 1

exercise a

$$Av = \Lambda v$$

First I am going to multiply A to the first column of Q

So $v_1 = [-a, 0, b]^T$ with eigenvalue 25.

This gets me the equation:

$$A \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} = 25 \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix}$$

Now I calculate $Av_1 =$

$$Av_1 = \begin{pmatrix} -2 & 0 & -36 \\ 0 & -3 & 0 \\ -36 & 0 & -23 \end{pmatrix} \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix}$$

Now we multiply horizontally on the left, with vertically on the right. So for example the element on the top left will be $(-2)(-a)$ and then $0(0)$ and then $-36(b)$ and so on.

$$\begin{pmatrix} (-2)(-a) & +0(0) & +(-36)(b) \\ 0(-a) & +(-3)(0) & +0(b) \\ (-36)(-a) & +0(0) & +(-23)(b) \end{pmatrix} = \begin{pmatrix} 2a - 36b \\ 0 \\ 36a - 23b \end{pmatrix}$$

Now I equate this to $25v_1$:

$$\begin{pmatrix} 2a - 36b \\ 0 \\ 36a - 23b \end{pmatrix} = 25 \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} = \begin{pmatrix} -25a \\ 0 \\ 25b \end{pmatrix}$$

And now I can just solve the equations for a and b :

$$-2a - 36b = -25a \implies 27a - 36b = 0 \implies 27a = 36b \implies b = \frac{27}{36}a = \frac{3}{4}a$$

$$36a - 23b = 25b \implies 36a - 48b = 0 \implies 36a = 48b \implies b = \frac{36}{48}a = \frac{3}{4}a$$

Since Q orthogonally diagonalizes A , the following must be true

$$(-a)^2 + (0)^2 + (b)^2 = 1 \implies a^2 + b^2 = 1$$

I can enter the value of b I found above into this equation so we get:

$$a^2 + \left(\frac{3}{4}a\right)^2 = 1 \implies a^2 + \frac{9}{16}a^2 = 1 \implies \frac{25}{16}a^2 = 1 \implies a^2 = \frac{16}{25} \implies a = \frac{4}{5}$$

Then enter this value of a into the equation for b :

$$b = \frac{3}{4}a = \frac{3}{4} \cdot \frac{4}{5} = \frac{12}{20} = \frac{3}{5}$$

$$\boxed{a = \frac{4}{5}, b = \frac{3}{5}}$$

exercise b

...

exercise c

...

exercise d

...

exercise e

...

Problem 2

exercise a

...

exercise b

...

exercise c

...

exercise d

...

exercise e

...

Problem 3

exercise a

...

exercise b

...

Problem 4

exercise a

...

exercise b

...

Problem 5

...

...